Sakala Grievance AI POC: Scope and 48-hour Plan

Oct 12, 2026 · @thulasi Ram

Summary

We demo an **AI grievance desk** for Sakala on Wednesday 14 Oct 2026 after 3 PM IST: a citizen speaks or types a complaint in Kannada or English, and the system routes it to the right department, service and officer in seconds, then shows the Chief Secretary where complaints pile up.

The story for Madam: data received Mon 12 Oct 3:48 PM, working POC in under 48 hours, built on the 9,999 real grievances Sakala shared. It goes beyond NL-to-SQL: auto-routing, voice intake, location-aware assignment and a live command dashboard, topped by an innovation layer that turns the complaint box into a governance nervous system.

This plan is for tonight's call to agree the scope. Proposed default: build the four must-haves in section 5, keep the rest as stretch.

Problem statements and action items from the call

Mohan named one core problem and several demo asks; everything else on the call was master data or open questions.

#	Problem statement (from the call)	What it means for the POC
P1	Citizens must manually pick the department when filing, which is cumbersome. AI should auto-classify each grievance to its department	Train on the 9,999 rows (Description to Department, Line Department, Service, Grievance Name, Subcategory)
P2	The classification must follow the real Government of Karnataka hierarchy (the 41-department link)	Build a master: Department > Line Department > Service > Service Type (Sakala or Non Sakala)
P3	Complaints arrive by voice too; AI should categorise straight from voice	Kannada/English speech to text, then the same classifier. Read out real rows from columns A and B
P4	Use location: pin code or GPS from the mobile app should route to the responsible engineer or officer	Map district, taluk and GP to the officer in "Pending with"

#	Problem statement (from the call)	What it means for the POC
P5	Show how many complaints come from each source (iPGRS website, mobile app, call centre and others)	Source chart and a Bangalore or district map with drill-down
P6	"AI cannot be only NL-to-SQL" and needs governance: roles and responsibilities as a single source of truth (SSOT), per our neural AI governance framework	Masters plus role mapping are the SSOT; AI never routes outside them

Action items

- [ ] Build Sakala masters: department hierarchy (41-department link), services, service types, district, taluk, GP (Pallavi, data team)
- [ ] Pull the service list and service types from the Sakala website (Pallavi)
- [ ] Ingest the sample data into a database (Ram)
- [ ] Build the POC: auto-classify, voice, geo routing, source chart, timeline (Pranay, with the team)
- [ ] Voice and R&D support (Malikarjuna)
- [ ] Everyone: think beyond NL-to-SQL, bring ideas to tonight's call
- [ ] Evening check-in with Mohan, Pallavi, Malikarjuna, Ram, Pranay to agree scope and timeline
- [ ] Present to the Chief Secretary on Wed 14 Oct after 3 PM

Roles above follow the transcript; Pallavi's ownership of the masters is inferred from the "41 departments link" exchange.

What the sample data supports

The data is strong for routing, voice and dashboards, and weak for SLA or breach claims. It is 9,999 iPGRS grievances from 3 to 9 Oct 2026, 17 columns, only 14% tagged Sakala (full review).

Area	What we have	Gap or caution
Text	Description, ~77% Kannada, 17% mixed, median 258 characters	788 rows are repeated camp templates; a few are junk text
Labels for training	Department (38), Line Department (195), Service (703), Grievance Name (838), Subcategory 1 and 2	Stray whitespace and near-duplicate labels; 775 rows filed as "General"

Area	What we have	Gap or caution
Location	District (31), Taluk (235), GP/Village/Hobli	GP blank in 35% of rows; no pin code or GPS
Workflow	Status, Officer Role (LMF 92%, L2 7%, L3 1%), Pending with / Closed By (2,956 officers)	Meaning of LMF, L2, L3 unconfirmed
Channel	9 sources: Praja Seva Abhiyana 41%, Citizen Service Centre 24%, iPGRS website 19%, Mobile App 9%, call centres 6%	One camp week dominates; not a statewide average
Time	Created Date only	No closed date, due date, SLA days or history, so no resolution time or breach demo
Privacy		434 texts hold phone numbers, 247 hold Aadhaar-like numbers; mask before display

One finding worth showing Madam: Mobile App complaints sit in "Registered & Sent for Scrutiny" 39% of the time, against nearly 0% for other channels, which is a manual triage bottleneck AI routing removes.

What AI can do beyond NL-to-SQL

Ten capabilities fit this data; the first five carry the demo and the rest are stretch or next phase. Sorted by demo value for the Chief Secretary.

#	Capability	What Madam sees	Data support	48h effort
1	**Auto-routing classifier**	Paste or speak a complaint; AI returns Department, Line Department, Service, Grievance Name, Sakala or not, with a confidence score and a one-line reason	Strong (9,999 labelled rows)	Medium: LLM with few-shot examples from similar past grievances, constrained to the master list

#	Capability	What Madam sees	Data support	48h effort
2	**Voice intake in Kannada and English**	A citizen speaks; transcript, English summary and routing appear live	Text only; we read real rows aloud as the call proposed	Medium: speech-to-text (ElevenLabs or similar) feeding #1
3	**Location-aware assignment**	Mobile app pin or pin code lands on a map and names the officer ("Pending with") for that taluk and department	District and taluk clean; no GPS or pin code	Medium: taluk centroid lookup plus officer mapping from the data; pin code assumed
4	**Command dashboard**	Complaints by source, department, district, status, officer load; Mobile App scrutiny bottleneck	Strong	Low to medium
5	**PII masking**	Phone and Aadhaar numbers hidden on every screen	Strong (681 rows affected)	Low: regex plus review
6	**Duplicate and hotspot clustering**	Near-identical complaints (say, drinking water in one taluk) grouped as one issue with a count	Strong (788 repeated rows)	Medium
7	**Ask the data (NL-to-SQL plus narrative)**	Madam asks "which taluk has the most pending housing complaints?" and gets a chart and a sentence	Strong	Low to medium
8	**Officer reply drafting**	AI drafts a Kannada or English reply and next action for the officer	Weak: no resolution text	Low as a mock-up
9	**Label clean-up and "General" rescue**	775 "General" rows re-routed to a real category	Strong	Low, falls out of #1

#	Capability	What Madam sees	Data support	48h effort
10	**Breach risk and auto-compensation**	Days left on SLA, overdue alerts, Rs 20/day compensation	Not supported: no due or closed date	Next phase, once Sakala shares SLA data

Guardrails that follow the SSOT point from the call: the classifier may only pick values that exist in the masters, low-confidence cases go to a human, and every routing shows its reason so officers can correct it.

Innovation layer: what makes the Chief Secretary remember it

Routing proves the AI works; these ideas show what Karnataka could do with it. The pitch line: "We turn Karnataka's complaint box into a governance nervous system." Each idea carries a tag for what we can show on 14 Oct.

#	Idea	What Madam sees	Demo on 14 Oct
11	**CS Monday brief**	An auto-written one-page brief in Kannada and English: top issues, worst taluks, masked citizen quotes, three decisions needed this week	Live, on real data
12	**Fix the system, not the case**	AI traces clusters of complaints to a root cause (for example a khata or mutation backlog in one office) and proposes the one process fix or order that prevents them	Live on real data; cause labelled as AI-inferred
13	**Vulnerability priority**	Complaints from elderly, disabled or widowed citizens, or about missing pension or rations, jump the queue	Live; falls out of the classifier
14	**AI calls the citizen back**	After closure, an AI voice calls in Kannada: "Was your problem solved?" A "no" reopens the case. Today only 64 of about 3,500 closures have citizen feedback	Live call to a team phone (Malikarjuna, voice)

#	Idea	What Madam sees	Demo on 14 Oct
15	**Grievance-to-Sakala bridge**	Flags complaints that are really a delayed Sakala service (217 Sakala-tagged "pension not received" rows) and the compensation they would earn	Dashboard tile; amount illustrative
16	**Citizens as sensors**	A spike in one topic and taluk raises an early warning to the DC before it becomes news	Vision slide; one week of data cannot prove trends
17	**From complaint to entitlement**	A pension complaint also lists other schemes the citizen qualifies for and starts those applications	Vision slide; needs scheme eligibility rules
18	**Closure quality auditor**	AI compares the complaint with the officer's closure note and flags closures that look fake	Next phase; needs closure remarks

Proposed POC scope and demo storyline

Must-have (commit tonight): capabilities 1 to 5 above (auto-routing, voice intake, location-aware assignment, command dashboard, PII masking), plus 11 CS Monday brief and 13 vulnerability priority as the innovation layer.

Stretch (only if must-haves are done by Tue night): 6 hotspot clustering, 12 fix the system, 14 AI callback, 15 Grievance-to-Sakala tile, then 7 ask-the-data and 9 "General" rescue.

Out of scope, shown as "next phase" slide: 16 citizens as sensors, 17 complaint to entitlement, 18 closure auditor, 8 reply drafting, 10 breach risk and compensation, live integration with iPGRS or Sakala systems.

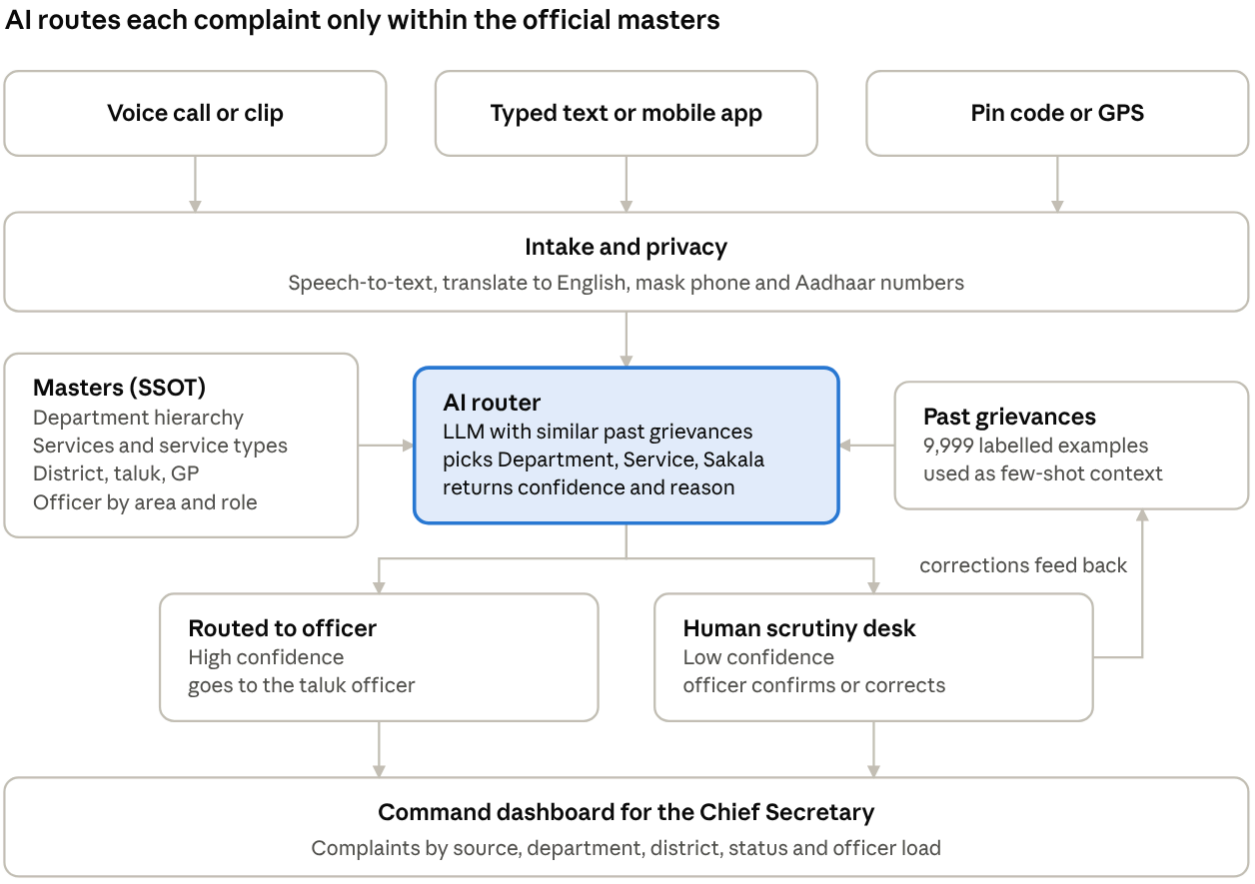
Demo storyline (about 10 minutes):

1. The problem in one slide: 9,999 complaints in one week, 9 channels, manual department selection, 775 filed as "General", Mobile App complaints stuck in scrutiny.
2. Voice: a team member reads a real Kannada complaint from the sample aloud. Transcript and English summary appear.

3. Routing: AI names Department, Line Department, Service and Sakala or not, with confidence and reason. Show one case where it disagrees with the original label and is arguably right.
4. Location: the complaint drops on a Bengaluru or district map and names the responsible officer for that taluk.
5. Dashboard: complaints by source, department, district and status; officer load; the scrutiny bottleneck.
6. Accuracy slide: tested on held-out rows, X% correct at department level and Y% at service level (numbers filled after Tuesday's test run).
7. Governance: masters as single source of truth, PII masked, human in the loop for low confidence.
8. Madam's own brief: hand her a Monday brief generated from this week's data, with one root-cause fix and, if ready, a live Kannada callback to a citizen. Close on the vision slides (citizens as sensors, complaint to entitlement) and what we need from Sakala for SLA breach alerts and compensation.

Architecture and governance guardrails

One pipeline serves every channel, and the masters act as the single source of truth the AI cannot step outside.



POC architecture · intake to dashboard

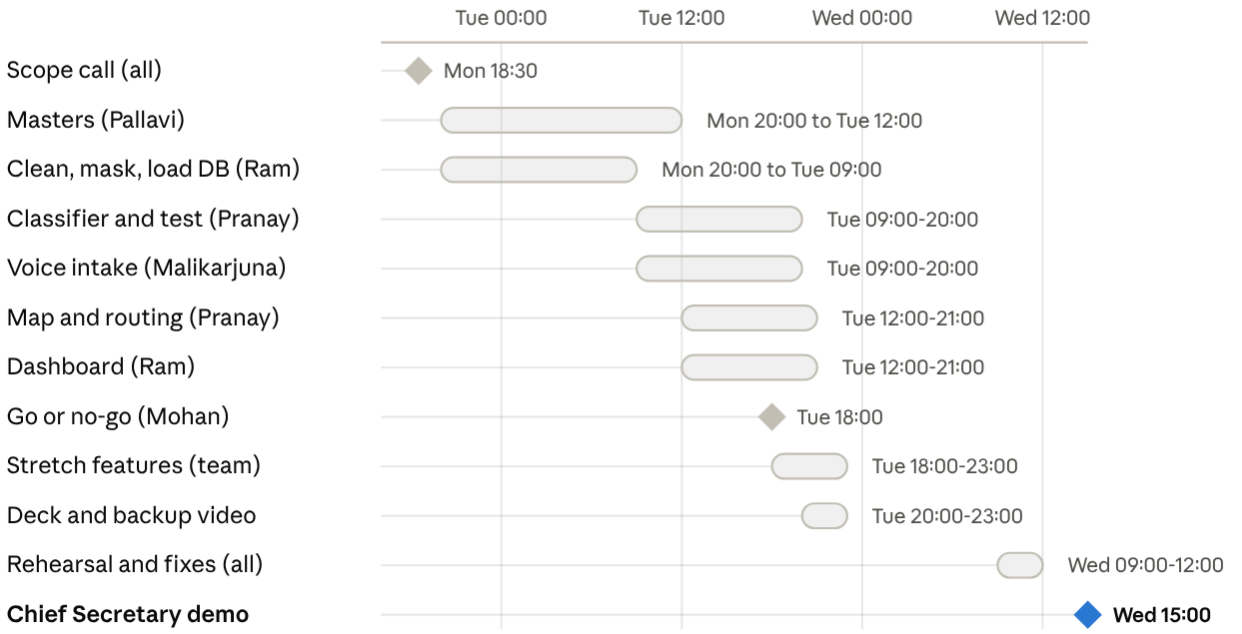
Suggested stack for 48 hours: Postgres for the cleaned data and masters, a Python API, Claude for classification and summaries, a Kannada-capable speech-to-text service, and a simple web app with a map (Leaflet) and charts. Data stays masked from ingest onward.

48-hour timeline and owners

The clock started Mon 12 Oct at 3:48 PM; masters and clean data land by Tuesday morning so the four build streams run in parallel through Tuesday.

Build runs Monday night to Tuesday night, Wednesday morning is rehearsal

48-hour plan, 12 to 14 Oct 2026, times in IST



48-hour timeline · 12 workstreams and milestones

Owners follow the call: Ram on data, Malikarjuna on voice and R&D, Pranay leading the build with the team. Masters with Pallavi and the dashboard with Ram are my assumption; reassign tonight. Tuesday 6 PM is the go or no-go: anything not working by then is cut, and the backup video is recorded that night.

Open questions for the Sakala team

None of these block the Wednesday demo; we assume a default for each and say so on the slide.

- [ ] What do LMF, L2 and L3 mean, and when does a grievance move from one to the next? (Assumed: LMF is the first-level field officer, L2 and L3 are escalation levels.)
- [ ] How is a grievance classified today: by the citizen, the call-centre operator, or a scrutiny desk? Who corrects wrong routing?
- [ ] Is the department list the 41 departments in our link, or the 38 that appear in the data? Which master is authoritative?
- [ ] What is the SLA for a grievance (as opposed to a Sakala service), and where are due and closed dates stored?
- [ ] Can we get closed or resolved dates, status history and resolution text for the same rows?
- [ ] Can we get a sample of real Sakala application data (GSC number, service, designated officer, applied, due and delivered dates)?

- [ ] How is "Pending with" mapped to geography: is there an officer master by taluk, GP or pin code?
- [ ] Does the mobile app capture GPS or pin code today?
- [ ] Are call-centre complaints recorded as audio? Could we get a few anonymised clips?
- [ ] Data-handling rules: may the data leave the State Data Centre for a POC, and which cloud or on-prem LLM is acceptable?
- [ ] Why do Mobile App complaints wait in "Registered & Sent for Scrutiny" so often?

Risks and decisions for tonight's call

Decisions needed tonight

1. Agree the must-have list (capabilities 1 to 5, plus 11 and 13) and the stretch list.
2. Pick the LLM and speech provider: Claude for classification (five R&D logins available), ElevenLabs or another Kannada-capable speech-to-text for voice.
3. Hosting for the demo: a laptop or private cloud instance with masked data only.
4. Map focus: Bengaluru wards or one district (Mysuru has dense rows, including Piriyapatna camp data).
5. Who presents which part on Wednesday, and the Tuesday 6 PM go or no-go.

Risks and fallbacks

Risk	Fallback
Kannada speech-to-text quality is poor on dialects	Use clear read-aloud recordings; keep a typed-input path ready
Classifier accuracy low at the 703-service level	Lead with department and line department accuracy; show service as "suggested top 3"
Masters from the 41-department link and Sakala website not ready in time	Derive the hierarchy from the sample data itself (labels are already 4-level)
No pin code or GPS in data	Simulate with taluk centroids and state the assumption
PII leaks on screen	Mask at ingest, before anything reaches the app or the LLM
Live demo failure	Record a backup video of the full flow on Tuesday night